

EQ Credit: Enhanced Indoor Air Quality Strategies

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To promote occupants' comfort, well-being, and productivity by improving indoor air quality.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Comply with **3 strategies for 1 point or 6 strategies for 2 points**

Strategy 1. Entryway Systems

Install permanent entryway systems at least 10 feet (3 meters) long in the primary direction of travel to capture dirt and particulates entering the building at regularly used exterior entrances.

Acceptable entryway systems include permanently installed grates, grilles, slotted systems that allow for cleaning underneath, rollout mats, and any other materials manufactured as entryway systems with equivalent or better performance. Maintain all on a weekly basis.

Warehouses & Distribution Centers only

Entryway systems are not required at doors leading from the exterior to the loading dock or garage but must be installed between these spaces and adjacent office areas.

Healthcare only

In addition to the entryway system, provide pressurized entryway vestibules at high-volume building entrances.

Strategy 2. Interior Cross-Contamination Prevention

Sufficiently exhaust each space where hazardous gases or chemicals may be present or used (e.g., garages, housekeeping and laundry areas, copying and printing rooms), using the exhaust rates determined in EQ Prerequisite Minimum Indoor Air Quality Performance or a minimum of 0.50 cfm per square foot (2.54 l/s per square meter), to create negative pressure with respect to adjacent spaces when the doors to the room are closed. For each of these spaces, provide self-closing doors and deck-to-deck partitions or a hard-lid ceiling.

Strategy 3. Filtration of Outdoor Air

Each ventilation system that supplies outdoor air to occupied spaces must have particle filters or air-cleaning devices that meet one of the following filtration media requirements:

- ▶ minimum efficiency reporting value (MERV) of 13 or higher, in accordance with ASHRAE Standard 52.2-2017; or
- ▶ Equivalent filtration media class of ePM₁ 50% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance.

Replace all air filtration media after completion of construction and before occupancy.

Data Centers only

The above filtration media requirements are required only for ventilation systems serving regularly occupied spaces.

Strategy 4. Filtration of Recirculated Air

Each ventilation system that supplies recirculated air to occupied spaces must have particle filters or air-cleaning devices that meet one of the following filtration media requirements:

- ▶ Minimum efficiency reporting value (MERV) of 13 or higher, in accordance with ASHRAE Standard 52.2-2017; or
- ▶ Equivalent filtration media class of ePM₁ 50% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance.

Replace all air filtration media after completion of construction and before occupancy.

Data Centers only

The above filtration media requirements are required only for ventilation systems serving regularly occupied spaces.

Strategy 5. Increased Ventilation 15 Percent

Increase breathing zone outdoor air ventilation rates to 95% of all occupied spaces by at least 15% above the minimum rates as determined in EQ Prerequisite Minimum Indoor Air Quality Performance.

Strategy 6. Increased Ventilation 30 Percent

Increase breathing zone outdoor air ventilation rates to 95% of all occupied spaces by at least 30% above the minimum rates as determined in EQ Prerequisite Minimum Indoor Air Quality Performance.

Strategy 7. Operable Windows

75% of the regularly occupied spaces have operable windows that provide access to outdoor air. The windows must meet the opening size and location requirements of ASHRAE 62.1-2016 with addendum I, section 6.4.1.2.

Strategy 8. Engineered Natural Ventilation

Achieve Option 2. ASHRAE Engineered natural ventilation system compliance path under EQ prerequisite. Minimum Indoor Air Quality Performance.

Strategy 9. Carbon Dioxide Monitoring

Monitor CO₂ concentrations within all densely occupied spaces. CO₂ monitors must be between 3 and 6 feet (900 and 1 800 millimeters) above the floor. CO₂ monitors must have an audible or visual indicator or alert the building automation system if the sensed CO₂ concentration exceeds the setpoint by more than 10%. Calculate appropriate CO₂ setpoints using methods in ASHRAE 62.1-2016, Appendix D.

Strategy 10. Additional Source Control and Monitoring

For spaces where air contaminants are likely, evaluate potential sources of additional air contaminants besides CO₂. Develop and implement a materials-handling plan to reduce the likelihood of contaminant release. Install monitoring systems with sensors designed to detect the specific contaminants. An alarm must indicate any unusual or unsafe conditions.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The credit has been restructured to a list of 10 strategies. New strategies are included for (1) filtration of recirculated air, (2) increased ventilation at 15% above prerequisite levels, and (3) operable windows.

Referenced standards for Strategy 3. Filtration of outdoor air and Strategy 9. Carbon dioxide monitoring have been updated to recent versions as follows:

- ▶ ASHRAE 52.2-2010 → ASHRAE 52.2-2017
- ▶ EN 779-2002 → ISO 16890-2016 ASHRAE 62.1-2010 → ASHRAE 62.1-2016

The strategies for natural ventilation and mixed mode ventilation have been updated to align with EQ prerequisite Minimum indoor air quality performance to encouraging natural ventilation design beyond prescriptive values as defined in ASHRAE 62.1-2016 addendum L as *naturally ventilated engineered systems*. Strategy 5. Increased Ventilation 15 Percent and Strategy 6. Increased Ventilation 30 Percent may be attempted by projects with mechanical ventilation, natural ventilation, or both. Increased natural ventilation must be documented consistently with EQ prerequisite Minimum IAQ Performance (i.e., using 62.1-2016 with addendum I and the referenced sections for prescriptive or engineered natural ventilation compliance paths, and increasing the required outdoor air quantities by 15% or 30%, as appropriate).

The strategy for Exterior Contamination Prevention has been removed from this credit but may be pursued as an innovation credit.

Step-by-Step Guidance

Follow LEED v4 reference guide with the following modifications:

Step 1. Select strategies

Determine which strategies the project will pursue.

If pursuing strategy 1. Entryway Systems

Refer to LEED v4 reference guide, Option 1. Enhanced IAQ Strategies, Entryway systems

If pursuing strategy 2. Interior Cross-Contamination Prevention

Refer to LEED v4 reference guide, Option 1. Interior Cross-Contamination Prevention and use ASHRAE 62.1-2016, Table 5.5.1 that lists minimum separation distances for air intakes.

If pursuing strategy 3. Filtration of Outdoor Air

Follow LEED v4 guidance for specifying outdoor air filtration media meeting credit requirements for minimum efficiency reporting value (MERV) ratings per ASHRAE standard or equivalent filtration media class of ePM1 50% or higher per ISO 16890- 2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance instead of EN 779-2002- Class F7 filters or higher.

If pursuing strategy 4. Filtration of Recirculated Air

Identify all HVAC equipment that supplies recirculated air to occupied spaces and specify filtration media that meet the credit requirements for minimum efficiency reporting value (MERV) ratings or equivalent filtration media class.

If pursuing strategy 5. Increased Ventilation 15 Percent

Follow the EQ Prerequisite Minimum Indoor Air Quality Performance prerequisite's steps for mechanical ventilation systems to determine the amount of outdoor air that must be supplied by each ventilation system. To meet the credit requirements, the system must deliver 15% more outdoor air to 95 percent of the occupied spaces at all times the space is occupied. For multiple-zone recirculating systems, this will likely increase the required outdoor air intake for the system by more than 15%.

If pursuing strategy 6. Increased Ventilation 30 Percent

Follow the EQ Prerequisite Minimum Indoor Air Quality Performance prerequisite's steps for mechanical ventilation systems to determine the amount of outdoor air that must be supplied by each ventilation system. To meet the credit requirements, the system must deliver 30% more outdoor air to 95 percent of the occupied spaces at all times the space is occupied. For multiple-zone recirculating systems, this will likely increase the required outdoor air intake for the system by more than 30%.

If pursuing strategy 7. Operable Windows

Identify all regularly occupied spaces and collect the following information for each space:

- ▶ Location of openings (on one side, two opposite sides, or two adjacent sides),
- ▶ Size of the opening (openable area).

Review section 6.4.1.2 in ASHRAE 62.1-2015 addendum I for window size and location requirements and compare the results with the design. Revise if necessary, to ensure that all regularly occupied spaces meet the requirements.

If pursuing strategy 8. Engineered Natural Ventilation

This strategy is met if the project complies with natural ventilation Option 2 under the EQ Prerequisite Minimum Indoor Air Quality Performance.

If pursuing strategy 9. Carbon Dioxide Monitoring

Refer to LEED v4 reference guide, Option 2. Additional Enhanced IAQ Strategies, Carbon Dioxide CO₂) monitoring section. Use ASHRAE 62.1-2016, Appendix C for determining CO₂ concentration set-points.

Densely occupied spaces are defined as: an area with design occupant density of 25 people or more per 1,000 square feet (93 square meters).

If pursuing strategy 10. Additional Source Control and Monitoring

Refer to LEED v4 reference guide, Option 2. Additional Enhanced IAQ Strategies, Additional Source Control and Monitoring

Further Explanation

Exhaust Rates for Interior Cross Contamination Prevention

Follow LEED v4 reference guide for this section as well as ASHRAE 62.1-2016, Table 6-5 that lists minimum exhaust rates for spaces whose exhaust requirements exceed the 0.5 cfm per square foot rate.

Required Documentation

Refer to LEED v4 Reference Guide content, with the following additions:

- ▶ Strategy 4. Mechanical schedules highlighting MERV or filtration media class for all units that supply return air
- ▶ Strategy 6. Increased ventilation: confirmation (calculations are documented under EQ Prerequisite Minimum Indoor Air Quality Performance)
- ▶ Strategy 7. Diagrams or description of location of operable windows and opening size calculations
- ▶ Strategy 8. Natural ventilation confirmation (calculations are documented under EQ Prerequisite Minimum Indoor Air Quality Performance)

Exemplary Performance

Comply with 8 strategies.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies undertaken for improving indoor air quality such as having permanent entryway systems, specific filtration requirements as well as monitoring of contaminants can contribute to better indoor environmental quality performance for the occupants during the operations phase.